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MER-Tracker: Towards High-Speed 3D Point Tracking via Multi-View Event-RGB Hybrid Cameras

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Abstract

This paper proposes the first task for high-speed 3D point tracking using multi-view Event-RGB hybrid cameras. We design a cuboid observation device comprising 4 RGB cameras (30fps) and 2 Event cameras to synchronously capture high-speed motions, and propose MER-Tracker, a high–frame-rate 3D point-tracking network that fuses the complementary strengths of dual modalities. We first respectively extract 2D motion-change features from the RGB and Event modalities, then apply linear interpolation and anchor sampling to fuse the discrete RGB 3D features and continuous Event 3D features after 3D lifting, and finally employ a LoRA-tuned Transformer based on temporal correlationship to predict the high-frame-rate 3D point trajectories over fast motions, accomplishing high-speed 3D point tracking. To verify the effectiveness of our method, we construct both real-world and simulated high-speed motion datasets. Experiments on our datasets show that we achieve accurate high-speed 3D point tracking at high-frame-rate (150fps), outperforming state-of-the-art methods.

1. Introduction

3D point tracking [39] refers to the estimation of continuous, temporally coherent trajectories of arbitrary points in three-dimensional space from visual observations. Compared with 2D point tracking [3, 5–7, 14, 20], 3D point tracking records motion details from a more comprehensive spatial perspective, and therefore holds greater promise for applications in robotics[8, 11] and 3D reconstruction[19], etc. Recently, as 3D vision has evolved, 3D point tracking has seen remarkable gains. Monocular methods like SpatialTracker[46, 47] could recover 3D point trajectories using a single moving camera, while multi-view methods such as MVTracker[34] leverage multiple fixed viewpoints to capture trajectories with more complete coverage and fewer occlusions. However, most successes remain limited to ordinary low-speed motion. Faithfully recovering 3D point trajectories in truly high-speed phenomena (e.g. a running man, insect wingbeats, or a spinning rotator) remains challenging. Accurately reconstructing such motions could illuminate physical processes and enable new experimental observations, offering substantial scientific value.

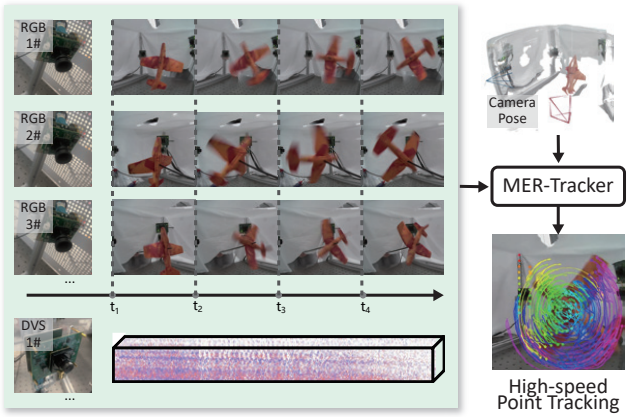


Figure 1. We propose MER-Tracker, a high-speed 3D point tracking method that predicts accurate high-frame-rate 3D trajectories using six different views of information, including four low-frame-rate, blurry RGB images and two continuous event streams.

A key bottleneck lies in sensing. Current 3D point tracking typically relies on optical RGB cameras to provide frames for building spatiotemporal models. However, commodity RGB sensors operate at relatively low frame rates (≈ 30fps), which leads to motion blur and large inter-frame gaps that miss crucial dynamics in fast motion. On the other hand, simply increasing frame rate with high-speed RGB

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cameras incurs substantial storage and bandwidth costs and demands strong illumination, imposing overly stringent experimental conditions.

Neuromorphic sensing offers a promising complement. Event cameras, also known as DVS (Dynamic Vision Sensor), are inspired by biological vision, providing microsecond-level temporal resolution and a vast dynamic range by asynchronously reporting brightness changes. Prior work has used event streams to mitigate camera-shake blur in static-scene reconstruction[33, 50]. However, because events encode temporal derivatives rather than absolute intensities, they capture edges and motion onsets well but lack dense textures and are largely insensitive to stationary regions. This raises a natural question: *Can we fuse the texture fidelity of RGB with the temporal acuity of event data to recover high–frame-rate 3D point trajectories for high-speed motions?*

We answer this question with MER-Tracker (Multi-view Event–RGB Tracker), a capturing system and learning method for high-speed 3D point tracking from multi-view RGB and event streams, as shown in Fig.1. Our design is driven by three challenges: (1) Building an Event–RGB fusion device that achieves multi-view, multi-modal spatiotemporal synchronization. (2) Extracting complementary 3D motion-dynamics features from two modalities and fusing them to yield a precise high-frame-rate 3D spatiotemporal representation. (3) Establishing a temporally continuous association between the high-frame-rate 3D spatiotemporal representation and each query point, guiding the Transformer to learn a generalizable 3D high-speed trajectory.

To this end, we first construct a cuboid capturing setup with 4 RGB and 2 event cameras that are tightly synchronized in time and calibrated in space. On top of this hardware, MER-Tracker introduces a dual-modal 2D feature extractor that firstly derives motion-change cues from RGB and DVS streams at their native rates, a heterogeneous 3D feature fusion that performs temporal interpolation and anchor sampling to aggregate these multi-view, multi-modal 3D motion features into compact spatiotemporal descriptors after 3D lifting, and a LoRA-tuned Transformer based on temporal correlations to effectively learn 3D point trajectories at high frame rates. Extensive experiments on both simulated and real datasets demonstrate that MER-Tracker produces accurate, temporally stable 3D point trajectories for high-speed motions. To our knowledge, we are the first to propose the systematic high-speed 3D point tracking task. We contribute datasets, methods, and a complete evaluation protocol to facilitate future research on this emerging task.

In summary, the main contributions of this work are:

- **Problem & System.** We introduce high-speed 3D point tracking via multi-view Event–RGB fusion, presenting a synchronized capture device and task formulation tailored to high-speed motion.
- **Method.** We propose MER-Tracker, featuring a dual-modal 2D feature extractor, a heterogeneous 3D feature fusion, and a temporal correlation Transformer that recovers dense, high-frame-rate 3D point trajectories.
- **Benchmarks & Evaluation.** We provide a real and a simulated dataset, along with evaluation metrics and a reproducible experimental framework, establishing a foundation for future work.

2. Related Work

2.1. Event-based Stereo Vision

Traditional RGB-based stereo vision has been extensively studied for decades, with major applications including novel view synthesis [1, 2, 10, 22, 27, 31, 37, 44, 51], point tracking [20, 42, 46, 52], etc. However, these frame-based methods often suffer from motion blur and limited temporal resolution in high-speed dynamic environments. Dynamic Vision Sensors (DVS, also called event cameras) operate asynchronously and output events triggered by pixel intensity changes. These sensors offer microsecond latency, high dynamic range, and low power consumption. Those advantages make them suitable for high-speed Stereo Vision tasks where conventional cameras fail [24].

Event cameras have achieved numerous innovative works in the field of 3D reconstruction. E2NeRF [33], EventNeRF [36], Ev-NeRF [18], and other works have explored the reconstruction of a 3D representation from a rapidly moving event camera. More recently, Evagaussians [50] and Eadeblur-gs [43] have fused event data and blurry frames to reconstruct high-quality images. However, these methods are primarily limited to static or quasi-static scenes, which restricts their applicability in dynamic environments. So E-4DGS [9] achieves dynamic reconstruction from multi-view event cameras, enabling more general and practical applications in dynamic 3D scene understanding and motion analysis. Dynamic 3D reconstruction results inherently rely on accurate point-level correspondences over time, as they depend on consistent point tracking throughout the scene. This close relationship motivates the use of point tracking techniques in event-based stereo vision.

2.2. 3D Point Tracking via Stereo Vision

While stereo vision aims to recover varying geometry, 3D point tracking addresses the complementary problem of maintaining consistent correspondences of scene points over time. Accurate point tracking is crucial for understanding complex motion patterns. This field has seen remarkable progress from recent methods [28, 29, 42, 46, 47, 49, 52]. SpatialTracker [42] lifts 2D points through a triplane representation into 3D and performs tracking in the 3D space. TAPIP3D [52] represents the input video as a sequence of multi-scale 2D feature maps enriched with 3D

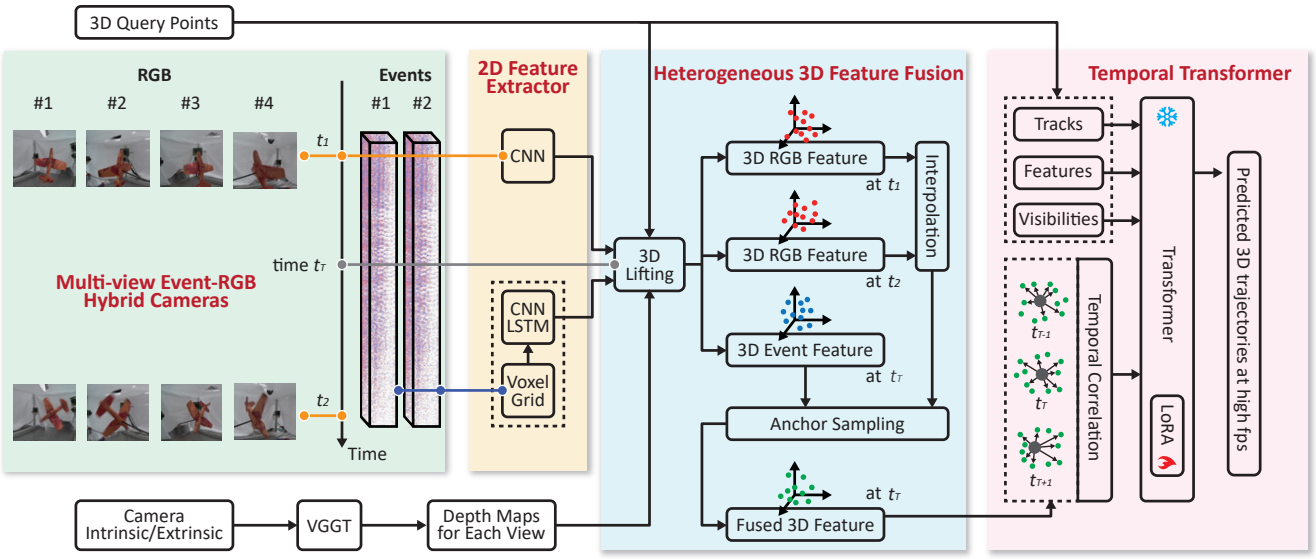


Figure 2. The main framework of our proposed MER-Tracker. MER-Tracker introduces a dual-modal 2D feature extractor that firstly derives motion-change cues from RGB and DVS streams at their native rates, a heterogeneous 3D feature fusion that performs temporal interpolation and anchor sampling to aggregate these multi-view, multi-modal 3D motion features into compact spatiotemporal descriptors after 3D lifting, and a LoRA-tuned Transformer based on temporal correlations to effectively learn 3D point trajectories at high frame rates.

coordinates. DELTA [29] proposes a coarse-to-fine trajectory estimation that manages to estimate dense point tracks across the whole image plane, instead of on a set of sparse locations. Those methods assume monocular video input.

In a different setting, MVTracker [34] aims to perform online 3D point tracking across multiple views, given a set of synchronized RGB frames with known camera parameters. Dynamic 3DGS [28] leverages 3D Gaussian reconstructions over time to track dense scene elements in 3D and perform accurate 2D and 3D point tracking with a multi-camera setup of 27 cameras in Panoptic Studio[23]. Such methods could achieve reliable point tracking results in both standard motion scenarios and synthetic scenarios. However, these methods struggle in high-speed scenes because conventional RGB cameras have limited frame rates. In contrast, our MER-Tracker is a framework that aims to complete point tracking in high-speed scenarios with the aid of DVS cameras.

3. Problem Statement

We aim to achieve high-speed 3D point tracking with accurate trajectories at a high frame rate. To reach this goal, we build a multi-view event-rgb hybrid stereo capturing device to capture the high-speed motion, and propose a novel method called MER-Tracker for point tracking based on continuous event streams with discrete RGB frames. The problem could be denoted as:

$$\textbf{MERTracker} : \{(\mathcal{I}_1, ..., \mathcal{I}_n, \mathcal{E}_1, ..., \mathcal{E}_m), t\} \rightarrow \hat{P}_t \quad (1)$$

where $\mathcal{I}_1, ..., \mathcal{I}_n$ are the RGB frames captured from n optical cameras, $\mathcal{E}_1, ..., \mathcal{E}_m$ are the event streams captured from m event cameras respectively, and $\hat{P}_t$ is the predicted trajectory of 3D points at target time t.

Specifically, for the n-th optical camera, the captured RGB frames $\mathcal{I}_n$ could be described as follows:

$$\mathcal{I}_n = \{I_n(t_i) | i \in \{1, .., F\}\}, \quad (2)$$

where $I_n(t_i)$ is the RGB image at the i-th frame time t_i, F is the total frames depending on the capturing frame rate (e.g. 30fps).

As for the m-th event camera, it captures continuous event streams $\mathcal{E}_m$:

$$\mathcal{E}_m = \{E_m(x_j, y_j, p_j, t_j) | j \in \{1, ..., N\}\}, \quad (3)$$

where $E_m(x_j, y_j, p_j, t_j)$ is the j-th captured event. It records the change in intensity of a specific pixel. (x_j, y_j) is the 2D coordinate of this pixel, $p_j \in \{+1, -1\}$ is the event polarity indicating whether the intensity at the corresponding pixel increases or decreases, t_j is the moment this event occurs, and N is the total number of events.

4. Methodology

4.1. Overview

To solve the problem of high-speed 3D point tracking with accurate trajectories, MER-Tracker first introduces a dual-modal 2D feature extractor that derives motion-change cues from RGB and DVS streams at their native rates, as detailed

in Sec. 4.2. Then, Sec. 4.3 performs temporal interpolation and depth lifting to construct modality-aligned 3D motion features, upon which anchor sampling is applied to aggregate the resulting multi-view, multi-modal representations into compact spatiotemporal descriptors. Finally, a temporal correlation Transformer with LoRA-tuned is adapted to learn full-trajectory 3D point dynamics at high frame rates in Sec. 4.4. The main framework of the MER-Tracker is shown in Fig. 2.

4.2. Dual-modal 2D Feature Extractor

As we can see, the optical camera provides discrete images $\mathcal{I}$ at a low frame rate, while the event camera records a continuous asynchronous event stream $\mathcal{E}$. Therefore, we first perform temporal alignment between the two modalities, followed by separate 2D feature extraction for each.

Event Voxel Grid for Alignment. For a time sequence of (t_{start}, t_{end}), the asynchronous event streams during this period are converted into the event voxel grid for subsequent processing and temporal alignment. We follow [53] to evenly populate the events in a sliding window into B (B=5) consecutive and non-overlapping bins, where each event contributes to its two adjacent bins.

RGB Feature Maps. For discrete RGB frames, we extract dense appearance features using a convolutional neural network $\phi(I_n(t_i))$. Following [20, 34, 52], we also compute the feature maps at 4 different scales.

Event Feature Maps. For continuous event streams, they exhibit (i) strong temporal ordering dependencies and (ii) low texture content per time step. Therefore, we build a CNN-LSTM network to extract temporally dependent, fine-grained 2D feature maps $\psi(E_m(t_{start}, t_{end}))$ from the event-camera voxel grid. Specifically, the CNN-LSTM consists of a three-layer convolution network and a long short-term memory network, as described in the implementation details. Note that the event features are projected to match the RGB feature dimension.

4.3. Heterogeneous 3D Feature Fusion

After extracting multi-view and multi-modal 2D feature maps, we need to map them into a unified 3D space to obtain a multi-view-consistent representation for spatial tracking.

3D Feature Lifting. Firstly, following [34], each valid pixel (u_x, u_y) on the 2d plane could be lifted into 3D using the depth and camera parameters:

$$\mathbf{x}^v = \mathbf{E}_t^{v^{-1}}(\mathbf{K}_t^{v^{-1}}(u_x, u_y, 1)^\top \cdot \mathbf{D}_t^v[u_y, u_x]), \tag{4}$$

where $\mathbf{E}$ and $\mathbf{K}$ are the camera extrinsics and intrinsics for v-th camera at frame time t, $\mathbf{D}$ is the depth map. Thus, we can obtain lifted points $\mathbf{x}$ from all RGB&Event cameras. Then, each of these lifted points $\mathbf{x}$ is assigned with its corresponding 2D feature maps:

$$\begin{aligned} \mathcal{X}_I^n(t) &= (\mathbf{x}^n, \phi(I_n(t))[u_y, u_x]), \\ \mathcal{X}_E^m(t) &= (\mathbf{x}^m, \psi(E_m(\tau_t)[u_y, u_x]), \end{aligned} \tag{5}$$

where $\mathcal{X}_I^n(t)$ is the 3D feature point cloud extracted from n-th RGB camera at frame time t, $\mathcal{X}_E^m(t)$ is the 3D feature point cloud extracted from m-th Event camera at frame time t, τ_t is a ultra-short time interval near t and $(u_x, u_y) \in \Omega_t$ means using the set of valid pixels in 2D plane.

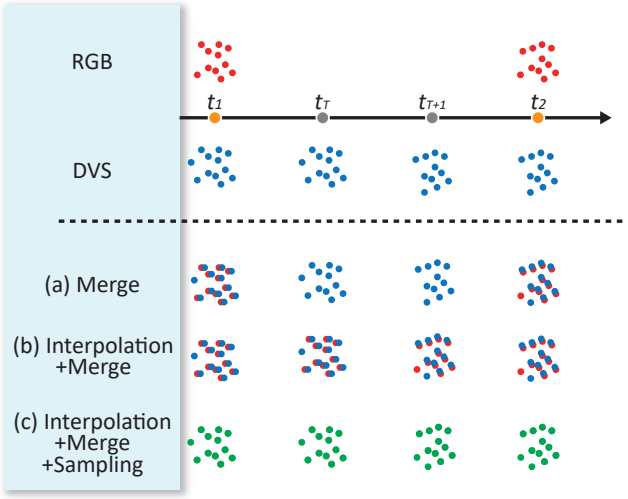


Figure 3. (a) Simply merging the point clouds from all viewpoints at each high-rate frame leads to temporal non-smoothness and spatial imbalance. (b) Linear interpolation yields a smoother temporal evolution of the feature point cloud, but it also introduces an excessive and spatially imbalanced distribution of points. (c) Our "interpolation + merge + sampling" method produces a fused feature point cloud with smoother variations, a more balanced distribution, and preserved original spatial structure.

However, as shown in Fig.3, to achieve fused 3D features at a high frame rate, simply concatenating the point clouds from all viewpoints at each high-rate frame leads to two critical issues:

- **Q1: Temporal non-smoothness**, when low-fps RGB frames are absent, the fused 3D features exhibit abrupt degradations;
- **Q2: Spatial imbalance**, simple merging overpopulates certain local regions with feature points while undersampling others.

Linear Interpolation for Smoothness. To solve the issue **Q1**, we need to supplement the missing RGB features at high frame rates. In general, trajectories of high-speed motion exhibit stronger temporal coherence. Accordingly, we approximate the evolution in the RGB feature space as linear and recover the missing high-frame-rate 3D feature point clouds via linear interpolation. Specifically, for a given RGB camera viewpoint, let t_1 and t_2 be two consec-

utive timestamps at a low frame rate (e.g., 30fps), and let $t_T \in (t_1, t_2)$ denote a target timestamp at a higher frame rate (e.g., 150fps). The supplemented 3D feature point cloud at time t_T could be computed as follows:

$$\mathcal{X}_I^n(t_T) = \alpha * \mathcal{X}_I^n(t_1) + (1-\alpha) * \mathcal{X}_I^n(t_2), \quad (6)$$

$$\alpha = (t_T - t_1)/(t_2 - t_1). \quad (7)$$

Anchor Sampling for Balance. To solve the issue **Q2**, we adopt a **merge-then-anchor-sampling** strategy: the two modalities are first aggregated, after which anchor-based resampling is applied. Using anchors yields a more balanced spatial distribution while preserving the original point-cloud geometry and structure. Specifically, for a target timestamp T, the fused 3D feature point of a total of n RGB cameras and m Event cameras could be obtained from:

$$\mathcal{X}(t_T) = \textbf{FPS}(\{\{\mathcal{X}_I^i(t_T)\}_{i\in n}, \{\mathcal{X}_I^j(t_T)\}_{j\in m}\}), \quad (8)$$

where **FPS** is the Farthest Point Sampling (FPS) method for anchor points extraction. As in PointNet++ [32], FPS iteratively selects points by maximizing the minimum distance between sampled points, yielding a balanced spatial distribution while preserving the original point-cloud geometry and structure.

4.4. Temporal Correlation Transformer

Temporal Correlation. After extracting the 3D features, the next step is to associate them with query points for trajectory prediction. Instead of leveraging spatial association modules (e.g., triplane projections or kNN neighborhood aggregation) to capture spatial correlations, we complement this by incorporating temporal correlation, forming a joint spatio-temporal nearest-neighbor correlation.

For each query point at the target timestamp t_T, we find the K nearest neighbors from adjacent frames:

$$\begin{aligned}\mathbf{C}_{t_T} = \{&(\mathbf{x}_k, f_k) \in \mathcal{N}_K(\hat{P}_{t_{T-1}}, \mathcal{X}(t_{T-1})),\\ &(\mathbf{x}_k, f_k) \in \mathcal{N}_K(\hat{P}_{t_T}, \mathcal{X}(t_T))\},\\ &(\mathbf{x}_k, f_k) \in \mathcal{N}_K(\hat{P}_{t_{T+1}}, \mathcal{X}(t_{T+1})),\end{aligned} \quad (9)$$

where $\mathcal{N}_K(\hat{P}_{t_T}, \mathcal{X}(t_T))$ finds the k-nearest neighbors around the current location estimate $\hat{P}_{t_T}$ in the fused 3D feature point cloud $\mathcal{X}(t_T)$. At high frame rates, motion is effectively "slow-motion," where each frame's position is tightly coupled with its immediate predecessors and successors. Modeling this temporal coherence yields stronger representations and preserves the continuity.

Lora Fine-tuned Transformer. We follow the work of [21, 34] to construct relationship tokens $G_{t_T} = \text{Enc}(\mathbf{C}_{t_T})$ and pass them to a transformer iteratively and get the full 3D trajectory at a high frame rate. To (i) leverage large-scale datasets for improved generalization and (ii) reduce training cost and dataset preparation overhead, we initialize the model with Transformer weights pretrained by MVTracker[34], and then apply parameter-efficient fine-tuning via LoRA(Low-Rank Adaptation)[15] on our self-made synthetic datasets.

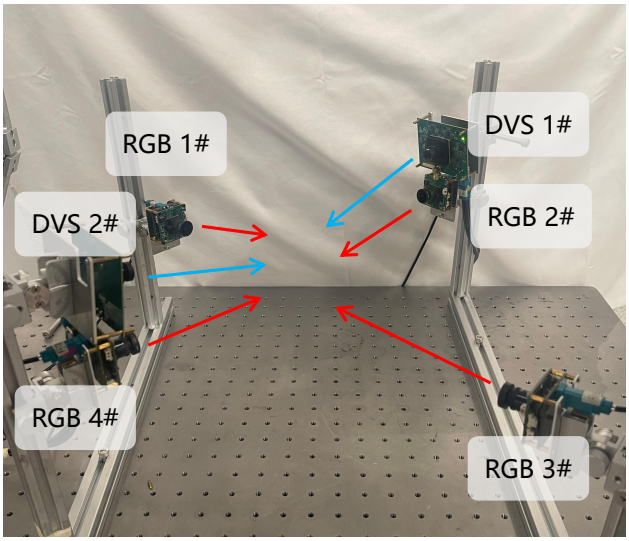


Figure 4. Our Multi-view Event-RGB Hybrid Camera System. We place 4 optical RGB cameras and 2 Event cameras (EVK4) to capture the high-speed object in the center of the device.

5. Dataset

5.1. Camera Prototype

To simultaneously acquire multi-view, high-resolution stereoscopic information and high–frame-rate texture dynamics, we built a carefully engineered experimental apparatus. We developed an event-guided, cube-shaped multi-view stereo capture rig consisting of 2 Event cameras and 4 optical RGB cameras. More precisely, our hybrid camera prototype realizes a cuboid observation geometry: 4 optical RGB cameras (30fps and 1080×720 resolution) and 2 event cameras (EVK4, microsecond temporal resolution and 1080×720 resolution are placed around the target, as shown in Fig.4. All camera data is transmitted via optical fiber to on-board processing chips for hardware time synchronization and encoding, then decoded and stored on a host workstation. During data collection, all cameras are time-synchronized to ensure a consistent depiction of motion across viewpoints. At present, the device serves small-scale testing in the laboratory. Owing to its scalability, it could later be installed on mocap frames or positioned outdoors to capture larger targets (e.g., human subjects and vehicles).

5.2. Real-world Object Dataset

As a first step, we collect real-scene data using the device we built. We select small, fast-moving, and representative

Table 1. Quantitative evaluation on different datasets. For the competitors, we use the model trained on 5K scenes from the MV-Kub dataset[34] and then evaluate them directly on the test sets of different datasets. In contrast, our method starts from mvtracker-initialized parameters and is fine-tuned on 70 training scenes from FMV-Kubric, after which it is directly evaluated on each test set.

Method	Train/Finetune	FMV-Kubric (30)				FMV-Panoptic (6)				Real Object (5)
		AJ ↑	δ_{avg} ↑	OA ↑	MTE ↓	AJ ↑	δ_{avg} ↑	OA ↑	MTE ↓	RMSE ↓
Tri-SpaTracker[46] + Traj.Inter	MV-Kub (5K)	56.1	66.8	85.4	2.8	55.2	70.5	72.3	9.2	0.338
MVTracker[34] + Traj.Inter	MV-Kub (5K)	61.9	72.7	87.9	2.3	62.5	83.2	75.8	7.2	0.324
MVTracker + Frame.Repeat	MV-Kub (5K)	60.5	71.0	87.5	2.5	58.3	72.8	73.5	8.6	0.332
MVTracker + Frame.Inter	MV-Kub (5K)	62.3	74.4	88.1	2.1	63.1	83.8	77.2	7.1	0.313
MVTracker + Frame.Inter + E2V	MV-Kub (5K)	63.5	75.2	88.3	2.0	65.2	84.0	82.6	6.9	0.307
Our MER-Tracker	FMV-Kubric (70)	**72.3**	**82.4**	**91.5**	**1.2**	**76.3**	**86.2**	**91.5**	**6.5**	**0.228**

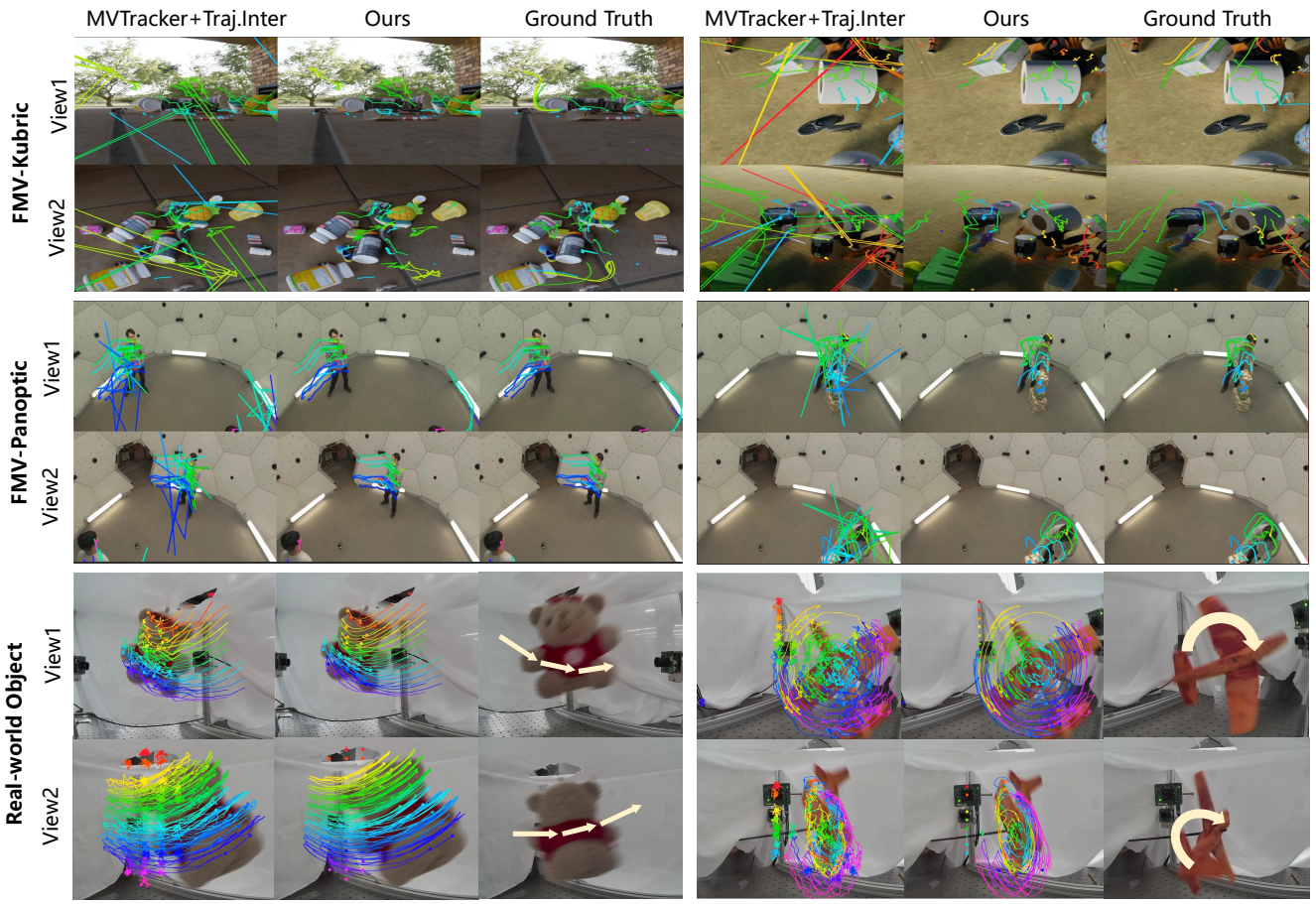


Figure 5. Qualitative comparison of multi-view 3D point tracking in high-speed scenes. We show the comparisons on three different datasets: FMV-Kubric, FMV-Panoptic, and Real-world Object. The SOTA method fails to predict accurate trajectories due to frame-missing and blur, while ours utilizes event streams for supplementary information. Note that we have no ground truth in the real-world object dataset, so we show the actual trend lines of the movement.

objects, such as a shaking cube, a spinning plane, and a swinging toy, totaling 5 typical scenes. During data acquisition, the capture device remains stationary, while the objects move within a constrained range centered on the device. Each scene recording provides 6 viewpoints: 4 RGB views containing discrete frame images at 30fps, and 2 event views storing event tuples (x, y, p, t) at microsecond temporal resolution. For 3D point tracking, i) the fixed camera extrinsics/intrinsics are acquired from spatial-temporal calibration [25], ii) the initial point clouds are obtained from VGGT[40] and utilized as query 3D points, iii) the depth maps for RGB cameras are also obtained from VGGT, and

the depth maps for Event cameras are unprojected-projected from RGB cameras.

5.3. FMV-Kubric Synthetic Dataset

Since our real-world rig does not include a depth camera (to the best of our knowledge, high-frame-rate depth sensors are scarce on the market), the real-world object dataset is not suitable as a standard benchmark for training and quantitative evaluation. We need a large-scale dataset with complete point-trajectory ground truth. Therefore, following MVTracker[34], we also build a virtual device in the Kubric physics simulator [12] comprising 4 low-FPS RGB cameras and 2 Event cameras arranged around the scene. Multiple objects undergo rapid free-fall from a fixed height at the scene center, and all cameras record synchronously. We synthesize a total of 100 scenes, 70 for training and 30 for testing. The camera parameters, 3D query points, depth maps, and ground truth trajectories at high frame rate are directly extracted from the Kubric simulator.

5.4. FMV-Panoptic Simulated Dataset

Because FMV-Kubric is fully synthetic, training and testing on it do not guarantee transferability to other environments, especially real-world scenes. Meanwhile, our real-world object dataset lacks a complete evaluation benchmark. We therefore construct a real-scene benchmark with exhaustive ground truth by synthetically modifying Panoptic[23]: we select six suitable viewpoints, temporally decimate four views (keep one out of every five frames) and apply motion blur, and convert the remaining two views into continuous event streams using v2e[16]. The resulting 4 low-frame-rate RGB sequences and 2 microsecond-temporal-resolution event streams serve as inputs, while the dataset's original 3D trajectories are used as surrogate high-frame-rate ground-truth trajectories.

6. Experiments

6.1. Experimental Setup

6.1.1. Competitors

Due to the relative lack of specific methods for 3D point tracking at high temporal resolution, we deploy the comparison using 2 kinds of two-stage prediction competitors. The first kind of competitor utilizes the low-frame-rate trajectories to directly predict high-frame-rate trajectories via linear interpolation on points. The second kind of competitor synthesizes a full high-frame-rate video by filling missing frames via "Repeat" (extending a video by repeating each frame n times), "Inter" (using the frame interpolation method such as SimpleFlow [38] and RIFE [17]), or "E2V"(converting DVS streams into intensity images via e2vid[35]); the resulting high-frame-rate frames are subsequently used to estimate high-frame-rate trajectories. For baseline SOTA multi-view 3D tracking methods, we select MVTracker[34] and triplane-expanded SpaTracker[34, 46].

6.1.2. Metrcis

For the synthetic dataset, which provides accurate high-frame-rate point ground truth, we adopt the same evaluation protocol as [34]: MTE (Median Trajectory Error) for visible points, quantifying spatial accuracy; (2) δ_{avg}, the average location accuracy over a set of threshold distances; (3) OA (Occlusion Accuracy), measuring the model's binary visibility prediction across views; and (4) AJ (Average Jaccard), which jointly evaluates occlusion and position accuracy. *For the real-world dataset*, since we do not have depth cameras to construct ground-truth point clouds, we design a surrogate evaluation metric. We first augment the original camera setup with an additional high-speed camera operating at 150fps, which records the motion of each frame from a novel viewpoint; the captured frames are then converted into depth maps using Depth-Anything[48]. For each predicted frame, we reproject the reconstructed point cloud into the high-speed camera view to obtain a rendered depth map. Finally, we compute a masked RMSE between the two depth maps. RMSE is defined as the square root of the mean squared error and is utilized frequently in recent works [30, 41, 45].

6.1.3. Implementation Details

We acquire 30fps, 720p RGB frames and 720p event streams with microsecond-level temporal resolution. Our goal is to reconstruct fast-motion 3D trajectories at 150fps (for real-world dataset). i) Stage I: 2D feature extraction. For RGB, CNN features are computed on each 30fps frame, with the input resolution downscaled to reduce computation. For Event, the stream is segmented at 150fps intervals, voxelized per segment, and processed by a CNN–LSTM. The CNN has three layers (64, 128, 256 channels; kernels 7×7, 3×3, 3×3), and the LSTM hidden dimension is 256. ii) Stage II: 3D feature fusion. RGB-derived 3D features are computed at 30fps and temporally upsampled to 150fps by linear interpolation. Event-derived spatiotemporal features, obtained per 150fps slice, are regarded as discrete 3D features at the same rate. We fuse the streams by selecting 3D keypoints with farthest point sampling (FPS), adopting a downsampling rate of 0.3. iii) Stage III: Trajectory prediction. We build tokens analogous to those in MVTracker[34] and use the identical Transformer design with an iterative training schedule. During fine-tuning, we load the MVTracker pretrained checkpoint, augment each Transformer layer with LoRA modules, train on our synthetic-dataset training set, and persist both the LoRA adapters and the Transformer weights for later evaluation. Our method is trained on 70 self-made FMV-Kubric synthetic scenes for 20k steps on 2 NVIDIA A6000 chips with 60GB GPU memory over 40 hours (batch size is 2). The model is im-

plemented in PyTorch.

6.2. Performance Comparisons

6.2.1. Comparison on FMV-Kubric Synthetic Dataset

We first evaluate on the FMV-Kubric synthetic dataset, which primarily comprises scenes of numerous objects rapidly falling from height in a simulation environment. We assess high-frame-rate trajectory prediction, with quantitative results reported in Table 1.Specifically, for the comparison with SOTA methods, (i) in the 1st-2nd row, we use only 30fps RGB frames in 4 views as input, with linear interpolation on trajectories for 150fps output; (ii) in the 3rd row, we expand the input RGB frames from 30fps to 150fps by repeating each frame 5 times; (iii) in the 4th row, we use RIFE[17] to interpolate intermediate frames for RGB inputs; (iv) in the 5th row, we use e2vid[35] to translate the 2 event views into images and combine them with RGB frames to form 6 views of input. We also show the qualitative performances in Fig. 5, showing the accurate prediction in high-speed synthetic environments.

6.2.2. Comparison on FMV-Panoptic Simulated Dataset

Next, we evaluate on the FMV-Panoptic simulated dataset comprising human actions such as passing a basketball, throwing a small ball, and placing a box. To emulate high-speed real-world motion, we subsample and blur 4 RGB views and simulate event data on 2 other views. Quantitative results are reported in Table 1. The qualitative performances are shown in Fig. 5, from which we could see that our method provides more accurate trajectory predictions with fewer perturbations in high-speed real-world simulated environments.

6.2.3. Comparison on Real-world Object Dataset

We also evaluate on our self-captured real-world object dataset. Using a custom multi-view, multi-modal optical rig to record small, fast-moving indoor objects, we assess the method's ability to predict and track fast-motion trajectories in real scenes. The quantitative results for RMSE are shown in Table 1, which show that ours is the best. We also present the qualitative performances in Fig. 5. It can be observed that our predicted trajectory best aligns with the object's true motion pattern.

6.3. Ablation Study

6.3.1. Contribution of Each Module

We conduct an ablation study to assess the contribution of each component in our MER-Tracker. As shown in Table 2, the results are evaluated by adding each module gradually: (i) The baseline results are shown in the first row, which uses only RGB frames as input to MVTracker[34]. (ii) In the second row, we directly merge RGB 3D feature points with Event 3D feature points to compose the fused feature point cloud. (iii) In the third row, we incorporate the linear interpolation for RGB 3D feature points and merge them with Event 3D feature points for direct use. (iv) The fourth row indicates that we utilize the anchor sampling to downsample the merged dual-modality 3D feature point cloud. (v) The last row is our full model, which incorporates the temporal correlation Transformer. Performance in this row achieves the best performance, validating the effectiveness of our design for the event-guided high-speed 3D point tracking task.

Table 2. The contribution of each component. Evaluated on the FMV-Kubric synthetic dataset.

Methods	AJ↑
Baseline	61.9
+ directly 3D merging	65.8
+ 3D interpolation	68.3
+ 3D interpolation + anchor sampling	71.2
+ 3D interpolation + anchor sampling + temporal TF	**72.3**

Table 3. 3D tracking performance under different numbers of combined cameras. Evaluated on the FMV-Kubric synthetic dataset.

RGB.Num	DVS.Num	AJ↑	δ_{avg} ↑	OA ↑	MTE ↓
3	1	66.5	76.3	88.4	1.7
3	2	70.6	81.1	91.0	1.4
4	0	61.9	72.7	87.9	2.3
4	1	68.1	78.6	89.9	1.6
4	**2**	**72.3**	**82.4**	**91.5**	**1.2**

6.3.2. Ablation on Camera Numbers

In this part, we investigate the relationship between camera numbers and performance. As shown in Table 3, we present both quantitative comparisons. It demonstrates that increasing the number of event cameras and RGB cameras leads to finer trajectory reconstruction, and that once a basic number of RGB views is available, additional event cameras play an increasingly dominant role. This means that our method could be extended to more views in the future.

7. Conclusions

In this paper, we propose the first task for high-speed 3D point tracking using Event-RGB hybrid cameras. We design a cuboid observation device comprising 4 RGB cameras and 2 Event cameras to synchronously capture high-speed motions, and propose MER-Tracker, a high–frame-rate 3D point-tracking network that fuses the complementary strengths of dual modalities. To verify the effectiveness of our method, we construct both real-world and simulated high-speed motion datasets. Experiments on these datasets show that our method achieves accurate, high-speed 3D point tracking, outperforming state-of-the-art methods.

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